

## Amnesty Human Rights DAO – Milestone 2 External Audit Summary

**Project Catalyst ID 1300018**

**Milestone:** 2 – Tokenomics Modelling & Simulation

**Date of Review:** 06/11/2025

**Reviewer:** Cherese Eriepa

### 1. Scope of Audit

The external review covered all smart contracts and simulation artefacts delivered under Milestone 2:

- soulbound\_governance.ak – minting policy for governance SBT vaults
- oracle\_rewards.ak – oracle-authorized rewards validator
- Tokenomics Model and Simulation Outputs (v2 parameter sheet, action weights, inflation/deflation logic)
- Test suite evidence and execution-unit metrics recorded via Aiken framework
- Documentation in /docs/ and /audit/ folders of [github.com/Sugarlicks/AmnestyDAO-Token](https://github.com/Sugarlicks/AmnestyDAO-Token) (branch: add-contracts/smart-contracts)

### 2. Methodology

The audit followed a three-stage process:

1. **Static Analysis** – Aiken compile & lint checks for syntax, policy logic, and parameter safety.
2. **Functional Verification** – Re-execution of 10 unit tests and traces for expected PASS/FAIL behaviour.
3. **Security and Performance Review** – Cross-checking signature enforcement, vault datum ownership, leakage protection, and execution-unit efficiency against Cardano protocol limits.

### 3. Findings

| Category                 | Issue                                                                                  | Severity | Resolution |
|--------------------------|----------------------------------------------------------------------------------------|----------|------------|
| Logic flow               | None detected – all tests passed as expected.                                          | –        | N/A        |
| Signature & auth checks  | Verified valid treasury and oracle signatory conditions.                               | –        | N/A        |
| Token policy ID guard    | Confirmed validation fails for counterfeit IDs.                                        | –        | N/A        |
| Output routing / leakage | Confirmed funds locked to vault and recipient only.                                    | –        | N/A        |
| Scalability metrics      | ≤ 75 K mem / 23 M CPU (SBT); ≤ 135 K mem / 51 M CPU (rewards). Within protocol limits. | –        | N/A        |

**Summary:** No critical or major vulnerabilities were identified. Minor notes relate to optional parameter optimization and comment clarity; these have been resolved in commit [hash].

### 4. Reviewer Statement

“I have independently reviewed the smart contracts, test results, and simulation outputs for Amnesty Human Rights DAO Milestone 2. All core functions executed as intended

with no security breaches or unintended behaviour. The design is sound, efficient, and aligned with the tokenomics model described in the Milestone 2 Technical Report.”

- The codebase is clean and well-structured, making the tests easy to run
- Core token contract functionalities are implemented correctly
- For future improvements, consider adding more edge-case tests for token transfers and governance interactions
- Adding inline documentation/comments would help with future audits and developer onboarding

*Cherese Eriepa*

**Signed:** \_\_\_\_\_

**Name / Title:** Cherese Eriepa / Software Engineer / Operations Lead

**Date:** 06/11/2025

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